

$ExtAxiom(t, VIS, AR) \triangleq$

$\forall x \in Keys,$

$v \in Values :$

$T_reads_ext(t, x, v) \Rightarrow$

LET $vis_writes \triangleq$

$\{tp \in CommittedTxns :$

$\langle tp, t \rangle \in VIS \wedge LastWriteIndex(tp, x) > 0$

$\}$

IN $T_writes(MaxAR(vis_writes, AR), x, v)$

$TransVisAxiom(t, VIS) \triangleq$

$\forall tp, s \in CommittedTxns :$

$(\langle tp, s \rangle \in VIS \wedge \langle s, t \rangle \in VIS) \Rightarrow \langle tp, t \rangle \in VIS$

$PrefixAxiom(t, VIS, AR) \triangleq$

$\forall tp, s \in CommittedTxns :$

$(\langle tp, s \rangle \in AR \wedge \langle s, t \rangle \in VIS) \Rightarrow \langle tp, t \rangle \in VIS$

$NoConflictAxiom(t, VIS, AR) \triangleq$

$\forall tp \in CommittedTxns :$

$(WriteConflict(tp, t) \wedge \langle tp, t \rangle \in AR) \Rightarrow \langle tp, t \rangle \in VIS$

$TotalVisAxiom(t, VIS, AR) \triangleq$

$\forall tp \in CommittedTxns : \langle tp, t \rangle \in AR \Rightarrow \langle tp, t \rangle \in VIS$

$IsConsistentExecution \triangleq$

$\forall t \in CommittedTxns \setminus \{InitTx\} :$

LET $lvl \triangleq Workload[t].level$

$V \triangleq Derived_VIS$

$A \triangleq Derived_AR$

IN $\wedge lvl \in IsoLevels \Rightarrow (IntAxiom(t) \wedge ExtAxiom(t, V, A))$

$\wedge lvl \in \{“CC”, “PC”, “PSI”, “SI”, “SER”\} \Rightarrow TransVisAxiom(t, V)$

$\wedge lvl \in \{“PC”, “SI”, “SER”\} \Rightarrow PrefixAxiom(t, V, A)$

$\wedge lvl \in \{“PSI”, “SI”, “SER”\} \Rightarrow NoConflictAxiom(t, V, A)$

$\wedge lvl \in \{“SER”\} \Rightarrow TotalVisAxiom(t, V, A)$

$OperationalSafety \triangleq IsConsistentExecution$

$Spec \triangleq Init \wedge \Box[Next]_{all_vars}$